Year 8 SEAL Science — Energy Practice Test (Hard)

TEACHER MARKING GUIDE — not for distribution to students

Total 40 marks: Section A 12 × 1 mark (separate answer sheet), Section B 28 marks. Writing time 55 min. Marking conventions: one mark per idea or step; no half marks. Mark units once, at the final answer. Where a later part uses an earlier answer, mark it consequentially. This is a hard practice paper: about two-thirds of the marks need interpretation, explanation or evaluation, and there are no sentence starters.

Section A — Multiple-choice answer key (12 marks)

Key balance: A ×3, B ×3, C ×3, D ×3; no letter repeats on consecutive questions. Each wrong letter points to one misconception or slip, so a tally of wrong letters shows what to reteach.

Q	Key	Working for the key	Distractor diagnosis
1	D	elastic potential energy	A: the pole moves, so thinks its store is kinetic. B: confuses the pole with the vaulter's height. C: thinks the muscles' chemical energy is stored in the pole.
2	B	h = 540,000 ÷ (1,500 × 10) = 36 m	A: inverted the formula (mg ÷ GPE). C: used ½ from the KE formula (GPE = ½mgh). D: left out g.
3	C	KE ∝ v^2: speed × 3, so KE × 3^2 = 9	A: thinks KE is proportional to speed. B: thinks squaring means doubling (2 × 3). D: used the difference in speed (30 − 10).
4	A	W: 400 ÷ 1,000 = 40% (X 30%, Y 37.5%, Z 20%)	B: chose the most light out, not the fraction. C: chose the smallest wasted energy, not the fraction. D: found wasted ÷ total and chose the largest.
5	C	radiation: needs no particles, so the air is not heated	A: thinks heat must pass through touching air. B: thinks warm air carries the heat down (warm air rises). D: confuses insulation with a heat-transfer method.
6	A	cold air is denser, so it sinks and stays low	B: particles themselves shrink when cooled. C: cold as a substance. D: 'heat rises' taken to mean heat can only move up.
7	B	15,000 W × 4 h = 60 kWh; 60 × $0.25 = $15.00	A: forgot the time (15 × 0.25). C: used 25 instead of 0.25 (cents as dollars). D: did not convert Wh to kWh.
8	D	shiny surfaces are poor emitters and good reflectors, so less radiant heat reaches the ice	A: cold as a substance. B: wrong method (convection). C: shiny metal 'conducts the heat away' from a distance.
9	C	burning releases the chemical energy stored in the tinder and wood as heat and light	A: thinks the stick's motion becomes the flame directly. B: thinks heat is stored in the wood like a fuel. D: thinks the tiny ember supplies all the flame's energy (it only starts the reaction).
10	D	more repeats reduce the effect of random error (reaction time)	A: precision, but the 0.2 s spread from reaction time is far bigger than 0.01 s. B: cherry-picks data (bias). C: changes a variable that should be controlled (validity).
11	A	dimming lights is a choice people make	B: appliance change. C: building change. D: appliance change. Each confuses changing equipment with changing behaviour.
12	B	the mat returns elastic energy she put in; her muscles add energy from chemical stores	A: energy created. C: thinks the elastic store builds up from bounce to bounce with no new input. D: ignores the energy her muscles add each push.

Section B — Short-answer marking scheme (28 marks)

Accept any clearly correct alternative at Year 8 level. Rejected answers are listed under each part.

Question 1 — Launch coaster (6 marks)

a) (4 marks)

KE after launch = ½ × 4,000 × 30² = 1,800,000 J

GPE at top = 4,000 × 10 × 40 = 1,600,000 J

KE at top = 1,800,000 − 1,600,000 = 200,000 J

200,000 = ½ × 4,000 × v^2, so v^2 = 100 and v = 10 m/s

1 mark KE after launch 1,800,000 J. **1 mark** GPE at top 1,600,000 J. **1 mark** KE left at top 200,000 J (consequential). **1 mark** v = 10 m/s from correct rearranging (consequential; unit needed)

Consequential: each later step on earlier values.

Reject: about 7.1 m/s from leaving out the ½ when rearranging (loses the last mark only); a speed found by subtracting heights or speeds directly (30 − 40) scores 0 for the last mark

b) (2 marks)

Highest possible hill = 1,800,000 ÷ (4,000 × 10) = 45 m; or GPE at 50 m = 2,000,000 J, more than the 1,800,000 J available.

1 mark a correct calculation: maximum height 45 m, or 2,000,000 J needed > 1,800,000 J supplied. **1 mark** judgement: the claim is wrong, the train would stop and roll back before the top; in reality friction and air resistance transform some energy into thermal and sound energy, so the real limit is even lower than 45 m

Reject: "Wrong, 50 m is too high" with no calculation scores the judgement mark only if a reason about energy is given; "energy is used up" is not a reason

Question 2 — Ice rink (5 marks)

a) (1 mark)

1 mark conduction

Reject: "convection"; "insulation"

b) (2 marks)

1 mark heat moves from the warmer water to the colder brine (not cold moving up): energy flows from hot to cold. **1 mark** particle link: the faster-vibrating water particles pass energy to neighbouring, slower concrete particles, which pass it on to the brine; the water particles slow down until the water freezes

Reject: "cold travels up"; "the brine makes the water cold" with no particle idea; "the particles get colder" without saying they lose energy or slow down

c) (2 marks)

1 mark each, for a change with the correct method named (max 2): doors or an airlock between foyer and rink, or sealing gaps (convection/draughts); a glass or plastic barrier between the stands and the ice (convection); bulk insulation in the walls and roof (conduction and convection); separate heating zones so warm air is kept in the stands (convection)

Reject: a change with the wrong method (e.g. sealing gaps reduces radiation); "turn the heating off" (behaviour, not a building change); a shiny roof lining, because the question is about warm air

Question 3 — Water slide pump (4 marks)

a) (2 marks)

X = 15,000 − 3,000 − 3,000 = 9,000 J

Overall efficiency = 9,000 ÷ 15,000 × 100 = 60%

1 mark X = 9,000 J. **1 mark** 60% (consequential on X)

Reject: 155% or 77.5% from adding or averaging the stage efficiencies

b) (2 marks)

Pump efficiency = 9,000 ÷ 12,000 = 75%

New useful energy = 15,000 × 0.95 × 0.75 = 10,687.5 J each second

Mass lifted = 10,687.5 ÷ (10 × 12) ≈ 89 kg each second, not 150 kg

Also: 150 kg × 10 × 12 = 18,000 J, more than the 15,000 J supplied

1 mark a correct calculation: about 89 kg per second, or 18,000 J needed for 150 kg. **1 mark** judgement with a reason: the claim is wrong; lifting 150 kg would need more energy than the motor takes in, which is impossible (efficiency cannot exceed 100%); the gain is only from 75 kg to about 89 kg

Reject: a verdict with no number; using 95% alone (15,000 × 0.95 ÷ 120 ≈ 119 kg) loses the calculation mark but can earn the judgement mark if the reason is correct

Question 4 — Stadium floodlights (5 marks)

a) (2 marks)

80 × 2,000 W = 160,000 W = 160 kW

160 kW × 1,200 h = 192,000 kWh

1 mark power × time set up for all 80 lights. **1 mark** 192,000 kWh

Reject: 192,000,000 kWh (W not converted); 2,400 kWh (one light only)

b) (2 marks)

LED: 80 × 1.1 kW × 1,200 h = 105,600 kWh

Saving = (192,000 − 105,600) × $0.25 = 86,400 × 0.25 = $21,600 a year

Payback = 130,000 ÷ 21,600 ≈ 6.0 years

1 mark yearly saving $21,600 (or both yearly costs, $48,000 and $26,400). **1 mark** payback about 6 years (consequential)

Consequential: on part a).

Reject: 130,000 ÷ 48,000 (used the old cost, not the saving)

c) (1 mark)

1 mark direction plus reason: the real payback is shorter, because the current lamps last only 6,000 ÷ 1,200 = 5 years, so all 80 must be replaced every 5 years (80 × $450 = $36,000), while the LEDs need no replacement within 40 years. Guide value: about 4.5 years if the lamp cost is spread evenly ($7,200 a year, so 130,000 ÷ 28,800 ≈ 4.5 years); a number is not required

Reject: "LEDs are better for the environment" (not from the table); "shorter" with no reason; a saving with no direction stated

Question 5 — Fire saw (2 marks)

a) (2 marks)

1 mark friction from sawing transforms kinetic energy into thermal energy in the dust, so its temperature rises. **1 mark** when the sawing stops, thermal energy is transferred from the hot dust into the cooler wood and air, so the dust cools; heat must build up faster than it escapes for the dust to reach the temperature needed for an ember

Reject: "they get tired" or "they run out of energy" with no energy transfer; "the heat is used up"

Question 6 — Pull-back toy car (6 marks)

a) (2 marks)

Outlier: 2.05 m (4 clicks, trial 2): it is far below the other two trials (3.12 m and 3.20 m).

Average = (3.12 + 3.20) ÷ 2 = 3.16 m

1 mark 2.05 m identified. **1 mark** 3.16 m (outlier left out)

Reject: 2.79 m (outlier included) for the second mark

b) (2 marks)

1 mark the distance increases as the number of clicks increases, with data (e.g. from 0.40 m at 1 click to 4.62 m at 5 clicks). **1 mark** each extra click adds more distance than the one before (e.g. +0.69 m from 1 to 2 clicks but +1.46 m from 4 to 5 clicks), so the relationship is not a straight line; doubling from 2 to 4 clicks almost triples the distance (1.09 m to 3.16 m)

Reject: "more clicks, further" with no numbers (first mark only if data are quoted); "directly proportional"

c) (2 marks)

1 mark the surface is an uncontrolled variable: the rougher court causes more friction, so the car travels a shorter distance for the same number of clicks; differences are then not caused by the clicks alone (less valid), and repeats spread out more (could explain the outlier). **1 mark** fix: run every trial on the same surface (e.g. always the gym floor), or record the surface and repeat any trial run on the court

Reject: "be more careful"; "do more trials" (does not control the surface); a fix with no effect explained scores the second mark only

Mark summary

Section	Marks	Notes
A — multiple choice	12	Answer sheet; key A3 / B3 / C3 / D3
B Q1 — Launch coaster	6	a 4, b 2
B Q2 — Ice rink	5	a 1, b 2, c 2
B Q3 — Water slide pump	4	a 2, b 2
B Q4 — Stadium floodlights	5	a 2, b 2, c 1
B Q5 — Fire saw	2	a 2
B Q6 — Pull-back toy car	6	a 2, b 2, c 2
Total	40	

Test blueprint

Hard level for SEAL: two-thirds of the marks need interpretation, explanation or evaluation. Q1a is the unscaffolded multi-step calculation (KE, GPE, then rearranging KE = ½mv² for *v*). Q3b chains efficiencies. 55 minutes for 40 marks is about 1.4 minutes a mark; the slowest items are SB 1a, 3b and 4b.

Demand level	Marks	Share	Hard target	Items
L1 Recall	4	10%	10%	MC 1, 5, 11; SB 2a
L2 Routine procedure	10	25%	25%	MC 2, 3, 7; SB 3a, 4a, 4b, 6a (1 of 2)
L3 Interpret or explain	16	40%	40%	MC 4, 6, 8, 9, 10; SB 1a, 2b, 5a, 6a (1 of 2), 6b
L4 Evaluate or design	10	25%	25%	MC 12; SB 1b, 2c, 3b, 4c, 6c

SB 6a: identifying and justifying the outlier is L3; the average is L2.

Coverage by unit lesson

Unit lesson	Marks	Items
2 Forms of energy, KE and GPE	9	MC 1, 2, 3; SB Q1
3 Heat transfer	5	MC 5, 6; SB 2a, 2b
4 Transfers, efficiency, Sankey diagrams	6	MC 4, 12; SB Q3
5 First Nations fire knowledge	3	MC 9; SB Q5
6 Appliances, running cost, payback	6	MC 7; SB Q4
7 Household energy design	3	MC 8; SB 2c
8 Investigation skills	7	MC 10; SB Q6
9 Energy audit	1	MC 11